Smaller explicit Diophantine equations for Con(ZFC)

Graph energies, shared digit bounds, and Pell centers

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Abstract

We construct several explicit integer-coefficient polynomials whose unsolvability, in their specified domains, is equivalent in ZFC to the syntactic consistency of ZFC. The main conceptual improvement is a graph compiler that checks the coefficient support actually used by a graph energy, rather than requiring all possible pairs of factor monomials to be distinct. Opposite occurrences are paired, compatible aliases are identified, and a four-term quartic removes certain equality witnesses. On exactly the source used by the previous unit-coefficient construction, the resulting quartic falls from 2,215 variables and 30,468 monomials to 1,137 variables and 23,105 monomials. With the newer signed source, we obtain unit-coefficient quartics in 951, 984, and 1,050 unrestricted integers. Their constant term is one and every other nonzero coefficient is also 1 or -1.

A second construction pools binomial low-digit bounds by their common denominator and combines this with a semantic Pell anchor. It gives a quartic in 226 nonnegative integers, with 175 quadratic residuals and 10,314 monomials; a 227-variable companion has only 7,905 monomials. A third, smaller improvement gives two ten-positive-variable polynomials of degrees 323 and 318 with 87- and 88-operation evaluators. All equations are literal finite objects in the archive. The new algebraic and arithmetic arguments are proved here. Their consistency interpretation uses explicitly identified, pinned ProveIt calibration theorems. No global minimum, consistency proof, historical priority claim, or new end-to-end proof-assistant verification is asserted.

Suggested starting points. For coefficient simplicity, inspect the 984-variable quartic with every nonzero coefficient ± 1. For the direct proof-calculus route, inspect the 226-variable natural quartic and its complete residual appendix. For a short evaluator, inspect the ten-positive-variable formulas in Section 6.

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1 The results and their exact meaning

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$. Positive integers are written $\mathbb{Z}_{>0}$; they are never silently interchanged with $\mathbb{N}$ or $\mathbb{Z}$. Let $\mathrm{Ref}_{\mathsf{ZFC}}(p)$ be the pinned syntactic predicate saying that p codes a finite refutation in ZFC, and put

$$\mathrm{Con}(\mathsf{ZFC}) \ :\Longleftrightarrow\ \neg\exists p\ \mathrm{Ref}_{\mathsf{ZFC}}(p).$$

The target is a theorem *about a fixed polynomial*, not a proof of the truth of either side of that theorem.

Theorem 1.1 (Delivered consistency equations)**.** *For every equation in Table 1, the integer companion in Section 5, and the two compact equations in Section 6, let D be its stated coordinate domain and n its number of unknowns. The new ordinary proofs in this report, composed with the identified inherited calibration, give*

$$\mathsf{ZFC} \ \vdash\quad \big[\exists \mathbf{x} \in D^n\ P(\mathbf{x}) = 0\big] \ \longleftrightarrow\ \exists p\ \mathrm{Ref}_{\mathsf{ZFC}}(p).$$

Consequently ZFC proves

$$\boxed{\big[\textit{there is no zero of } P \textit{ in } D^n\big] \ \longleftrightarrow\ \mathrm{Con}(\mathsf{ZFC}).}$$

For the graph equations, the new part is stronger: it is an unconditional explicit bijection between their integer zero sets and the zero sets of the pinned signed source systems.

Equation and domain	Variables	Degree	Monomials	Height
Graph, fewest variables, $\mathbb{Z}$	951	4	56,916	1
Graph, balanced choice, $\mathbb{Z}$	984	4	30,076	1
Graph, sparse choice, $\mathbb{Z}$	1,050	4	21,766	1
Graph, matched older source, $\mathbb{Z}$	1,137	4	23,105	1
Direct calculus, $\mathbb{N}$	226	4	10,314	87,839,583
Direct calculus, sparse, $\mathbb{N}$	227	4	7,905	87,839,583

Table 1: Literal collected polynomials. Height is the largest absolute value of a collected coefficient. All graph polynomials have constant one.

The additional compact equations have ten strictly positive unknowns, degrees 323 and 318, and 87 and 88 live binary arithmetic operations, respectively. Each has the same four fixed large coefficients reconstructed from a complete finite recipe; these are not extra variables. Its support is not expanded in this release.

1.1 Which meaning of "simplest" improves?

There is no established global simplicity minimum. The repository already contains quartics with fewer unrestricted integer variables when large coefficients are allowed, and higher-degree equations with fewer variables still. We therefore compare within the relevant families and retain the actual tradeoffs.

The same-source graph reduction saves 1,078 variables, about 48.7 percent, and 7,363 monomials, about 24.2 percent. This isolates the new compiler from the additional gain obtained by switching to the newer signed324 source. The 226-variable natural equation improves all displayed size measures over natural233. Against natural234, the 227-variable alternative has fewer variables and terms but a slightly larger coefficient height.

Matched comparison	Variables	Monomials	Height
Package 127, signed442 source	2,215	30,468	1
New compiler, same signed442 source	1,137	23,105	1
Package 129, natural233	233	50,762	88,674,690
New direct natural226	226	10,314	87,839,583
Package 129, natural234	234	9,019	87,741,134
New sparse natural227	227	7,905	87,839,583

Table 2: Comparisons to the inspected repository snapshot. All rows have degree four. The final sparse comparison slightly increases coefficient height; it is not a simultaneous improvement in every measure.

The comparison snapshot is ProveIt commit

`1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`.

The comparison includes the seven recently integrated packages 127–133. Relevant references are listed at the end.

2 A graph polynomial with exactly the source zero set

This section is independent of Pell equations, Turing machines, and set theory. It applies to a finite system of quadratic equations over $\mathbb{Z}$. The inherited signed324 source is a system

$$R_1(\mathbf{z}) = \cdots = R_{208}(\mathbf{z}) = 0, \qquad \mathbf{z} \in \mathbb{Z}^{324},$$

whose literal rows are retained as `graph/source324/equation.json`. Its consistency calibration is discussed in Section 7.

2.1 Normalization with a unique integral extension

First replace nonunit integer coefficients by explicitly defined constant and scale coordinates. For each positive coefficient $c > 1$, recursively include the constants needed for

$$c = 2\lfloor c/2 \rfloor + (c \bmod 2).$$

The corresponding equation uses two occurrences of the half-constant, each with coefficient -1, rather than a coefficient -2. For cz_iz_j, use a scale coordinate $s_{c,i} = cz_i$, then the product $s_{c,i}z_j$. Linear and constant terms are handled by the same fixed constants. The definitions are triangular and monic.

The result is a list of signed term occurrences

$$S_r = \sum_{j=1}^{s_r} \epsilon_{rj} m_{rj}, \qquad \epsilon_{rj} \in \{-1, 1\}, \quad \deg m_{rj} \le 2.$$

Every original tuple has a unique extension to the new coordinates; all added defining rows then vanish, and the other rows equal the original residuals. For signed324, normalization gives 506 coordinates, 390 rows, and 1,748 signed occurrences. For signed442 it gives 650 coordinates, 498 rows, and 1,969 occurrences. Occurrences, rather than only collected monomials, matter at this stage.

2.2 Incidence energy and integral tree flows

Let a finite directed graph have an integer-valued polynomial label t_e on every edge. Define its divergence at vertex v by

$$d_v = \sum_{\mathrm{tail}(e)=v} t_e - \sum_{\mathrm{head}(e)=v} t_e, \qquad E_G = \frac{1}{2}\sum_v d_v^2.$$

The polynomial E_G has integer coefficients: an edge square occurs at its two endpoints, while every cross term in a squared divergence has an even coefficient. Parallel edges do not invalidate integrality. They can invalidate a particular height-one bound, which is why the collected coefficient check is retained.

Lemma 2.1 (Tree completion)**.** *Partition the vertices into nonempty blocks. Inside every block the free edges form a spanning tree and carry distinct unrestricted integer variables. All other edge values are prescribed. There is an internal flow with every divergence zero exactly when the prescribed divergence sum in each block is zero. When it exists it is unique and integral.*

Proof. Necessity follows by summing divergences within a block; internal edges cancel. Root the internal tree. At each leaf, its one remaining tree-edge value is determined, with coefficient 1 or -1, by the other incident values. Eliminate the leaf and proceed. The final root equation is precisely the block sum. This proves existence, uniqueness and integrality without division. □

Use one block for each normalized row and one common hub block. Place prescribed signed monomial edges so that the row-block sums are exactly the S_r. The hub sum is then $-\sum_r S_r$. Introduce free tree flows inside the blocks. By Lemma 2.1,

$$\exists \text{ integral free flows } E_G = 0 \quad \Longleftrightarrow \quad S_r = 0 \text{ for every } r.$$

The graph need not be a tree globally.

2.3 Pairing opposite occurrences

Suppose the same monomial occurs with sign $+1$ in row r and sign -1 in row s. Instead of two edges to the common hub, use one edge from block r to block s, with that monomial as label. Each row sum is unchanged, and the two hub contributions cancel. This removes a prescribed edge and its associated factor copy.

The delivered deterministic layouts match 137 opposite pairs on the signed324 source and 141 on signed442. The initial layout avoids parallel prescribed edges, giving an easy initial coefficient bound. This is a construction choice, not a necessary hypothesis of the final zero-set theorem. The independent verifier checks the final block sums directly, so it need not trust the matching algorithm.

2.4 Aliases and a unit path

An *alias* is a coordinate intended to equal a particular normalized source coordinate. Initially, each prescribed monomial receives private factor copies, with some shared second factors. Linear and constant terms are made homogeneous of degree two using aliases of the number one. Thus prescribed edge labels have degree two and free edge labels have degree one. The copied source value of an alias is recorded explicitly.

For a class of $k \geq 3$ ordinary aliases define

$$C(a_1, \ldots, a_k) = \sum_{j=1}^{k} a_j^2 - \sum_{j=1}^{k} a_j a_{j+1}, \qquad a_{k+1} = a_1.$$

Since $2C = \sum_j (a_j - a_{j+1})^2$, it is nonnegative over $\mathbb{R}$ and vanishes exactly when all its aliases agree. A singleton needs no penalty. A two-element class can initially be extended by one dummy alias to obtain a three-cycle.

For the unit aliases $u_1, \dots, u_k$, $k \geq 2$, use

$$\mathcal{U}(\mathbf{u}) = 1 + \sum_{j=1}^{k} u_j^2 - u_1 - u_k - \sum_{j=1}^{k-1} u_j u_{j+1}.$$

Twice this polynomial is

$$(u_1 - 1)^2 + \sum_{j=1}^{k-1} (u_j - u_{j+1})^2 + (u_k - 1)^2.$$

It therefore vanishes precisely when all unit aliases equal one. Ordinary classes, unit aliases, and free flows are disjoint.

3 Why checking actual support permits much more compression

3.1 A global factor restriction is unnecessarily strong

The earlier compiler guaranteed that every product of two prescribed monomials uniquely identified its two occurrences. A sufficient condition was the absence of a four-cycle in a bipartite factor graph. But the incidence energy contains only products from edges sharing a vertex. Products of nonadjacent edge labels never appear.

For a concrete example, put the edge labels ab, ad, cd, cb, in that order, on a directed four-cycle. Its energy is

$$\begin{aligned} E = {} & a^2b^2 + a^2d^2 + c^2d^2 + b^2c^2 \\ & - a^2bd - acd^2 - bc^2d - ab^2c. \end{aligned}$$

Every nonzero coefficient is 1 or -1. Yet the factor graph has a four-cycle, and $(ab)(cd) = (ad)(cb)$. Those two equal products come from opposite edge pairs, so neither occurs in the energy. This is the precise gap exploited by the new compiler.

Theorem 3.1 (Certified alias identification)**.** *Start with a graph construction with complete ordinary alias classes and a unit class. Identify any coordinates only within the same class, leaving flow coordinates distinct. Substitute those identifications into the graph energy and rebuild valid equality penalties on the surviving aliases: singleton classes need none, two-element classes use a dummy three-cycle or Lemma 3.2, and the unit path retains at least two aliases. If the complete collected polynomial has coefficients in $\{-1, 0, 1\}$, its coefficient height is one and its integer zero set remains in explicit bijection with the original source zero set.*

Proof. Every old zero already gives the same value to the aliases being identified. Conversely, any tuple for the identified coordinates expands by copying each coordinate into all of its old aliases. At a zero of the new polynomial, nonnegativity makes every component vanish. The rebuilt ordinary penalties enforce the source labels, the unit path enforces one, and the graph divergences vanish. The row-block sums consequently give every normalized equation. Tree completion supplies the unique integral flows in the forward direction. All normalizing constants and scales were also uniquely determined.

This is a zero-set argument; rebuilding a cycle need not give the same polynomial as substituting into the old cycle. The energy substitution and the rebuilt penalties must be checked separately. The height

claim is exactly the finite collected-coefficient condition in the statement. It does not require global monomial-product uniqueness. □

The implemented search tries identifications of equal-labelled aliases and updates every affected energy coefficient with exact integers. An identification is accepted only when no collected coefficient leaves $\{-1, 0, 1\}$. There is no numerical sampling and no optimization oracle in the definition of the final polynomial. A fixed profile and a complete literal certificate are supplied; the verifier independently checks the resulting object from the original source.

3.2 A four-term equality removes a dummy coordinate

Lemma 3.2 (A unit-coefficient equality polynomial)**.** *The polynomial*

$$\boxed{\mathcal{Q}(a,b) = a^4 + b^4 - a^3b - ab^3 = (a-b)^2(a^2+ab+b^2)}$$

is nonnegative over $\mathbb{R}$, vanishes exactly on $a = b$, and has degree four and four nonzero coefficients, all of height one.

Proof. The second factor equals $(a + b/2)^2 + 3b^2/4$. It is positive away from $(0, 0)$, which already lies on the diagonal. The displayed factorization proves the assertions. □

For a two-alias class, replace its dummy three-cycle by $\mathcal{Q}(a, b)$. This saves one coordinate. Unlike a quadratic cycle, the new penalty can overlap the graph's quartic terms, so we accept it only after checking the full collected sum. The balanced equation saves 57 dummy coordinates this way; the minimum-variable equation saves 58, the sparse equation 59, and the matched older-source equation 67.

Proposition 3.3 (A narrowly defined local optimum)**.** *Among nonzero real-nonnegative polynomials in two variables with integer coefficient height one and zero set exactly $a = b$, degree four is minimal. Four nonzero monomials are also minimal.*

Proof. A globally nonnegative nonzero polynomial cannot have odd total degree: its highest nonzero homogeneous part changes sign under simultaneous negation. A polynomial of degree at most two vanishing on the entire diagonal and nonnegative everywhere must be $c(a-b)^2$, $c > 0$. This also follows by writing the polynomial in $a - b, b$; nonnegativity forces the coefficient of the first-order transverse term to vanish. Integrality gives $c \geq 1$, so its mixed coefficient has magnitude at least two. Thus degree below four is impossible.

After substituting $a = b = t$, the sum of the coefficients at each total degree must be zero. With coefficient height one, every nonempty degree group therefore has an even number of terms. With fewer than four terms a nonzero candidate has exactly two opposite same-degree monomials. On the positive quadrant their difference changes sign as a/b passes through one. It cannot be globally nonnegative. Lemma 3.2 attains both bounds. This proposition is about this local equality gadget, not global Diophantine simplicity. □

3.3 The final equation and an exact lift identity

Let $\mathcal{C}$ be the surviving ordinary cycles and $\mathcal{T}$ the selected two-alias classes. Each delivered graph polynomial is

$$\boxed{P_G = \frac{1}{2}\sum_v d_v^2 + \sum_{C\in\mathcal{C}} C + \sum_{(a,b)\in\mathcal{T}} \mathcal{Q}(a,b) + \mathcal{U}(\mathbf{u}) = 0.} \tag{1}$$

The fixed vertices, edges, edge labels and alias classes are completely listed in its certificate. The polynomial JSON also lists every collected coefficient directly. The factor $1/2$ is not a hidden rational coefficient in the collected polynomial: integrality and height one are both checked coefficient by coefficient.

There is an explicit polynomial lift L from the original source coordinates, obtained by computing its normalizing constants and scales, copying the alias values, setting the units to one and filling tree flows by leaf elimination. For arbitrary source tuples, not just zeros,

$$P_G(L(\mathbf{z})) = \frac{1}{2}\left(\sum_r R_r(\mathbf{z})^2 + \left(\sum_r R_r(\mathbf{z})\right)^2\right). \tag{2}$$

Indeed, only the block roots can retain divergence; they carry the individual residuals and their negative sum. All equality penalties vanish. In particular the right side is *not generally just* $\sum R_r^2$. The independent checker proves the root-divergence identities symbolically and checks the complete coefficient expansion.

4 Pooling low binomial digits in the direct calculus construction

The graph theorem is a general compiler improvement. The next construction changes the actual arithmetic certificate for a finite C222 derivation. It starts from package 129's fully specified 344-coordinate normalized natural graph, whose semantic interface and proofs are retained in the direct component. No K215 simulation and no Jones universal polynomial is instantiated on this route.

4.1 The inherited arithmetic interface

The source has a horizon $t \geq 16$, a proposed factorial F, and parameters x, y, with

$$R = 2 + t + x + y + F, \qquad K = 2R, \qquad N = B^K, \qquad q = N^K.$$

It also represents

$$V_0 = (F^2+1)^t, \quad D_t = q^t, \quad D_{t+1} = qD_t, \quad D_y = q^{y+1}, \quad D_x = q^{x+1}.$$

These variable powers are *semantic descriptions* of retained quadratic Pell circuits. They are not operations of the final polynomial. Their intended meaning is established in the inherited soundness order, before the factorial/binomial interpretation is used.

There are nine independent positive CRT representatives $a, b, c, d, S, S', u, v, C$, and a derived nonnegative constructor label J. The Cantor constructor is

$$I(a,b) = 2(a+b)(a+b+1) + 4b.$$

The actual label polynomial contains $I(a,b)$ and other nonnegative terms, so $J \geq 2a^2$. Eleven numerator requests have the following four common low-digit denominators:

Numerator requests	Denominator
Factorial binomial and C	D_t
a, b, c, d, J	D_{t+1}
S, S'	D_y
u, v	D_x

Originally each remainder r_i had a private nonnegative slack s_i, with $D_i = r_i + s_i + 1$. We pool these eleven bounds into four. The separate factorial digit-output condition $q = B + o + 1$ is retained.

Lemma 4.1 (Shared low-digit budget)**.** *Let $q \geq 2$, $k \geq 1$, and $n_1, \ldots, n_m \in \mathbb{N}$. Suppose*

$$\sum_{i=1}^{m} \binom{n_i}{j} \leq q - 1 \qquad (0 \leq j < k).$$

For the canonical low residues $r_i = \sum_{j=0}^{k-1} \binom{n_i}{j} q^j$, one has $\sum_i r_i \leq q^k - 1$. Hence a single nonnegative slack satisfies

$$q^k = \sum_i r_i + s + 1.$$

Conversely this one row, with arbitrary natural remainders, restores all old bounds by $s_i = s + \sum_{h \neq i} r_h$.

Proof. Sum the coefficient bounds against q^j and use $(q-1)\sum_{j<k} q^j = q^k - 1$. The restoration identity is literal and all its coefficients are nonnegative. □

Pooling strengthens the arbitrary-witness constraints. Separate bounds $r_i < q^k$ alone would not justify it. Completeness therefore needs a new uniform digit-margin estimate, proved next.

4.2 The actual radix has ample unused low-digit capacity

This subsection is only a constructive *completeness* argument. Start from a finite C222 refutation and choose the inherited true witnesses, so $F = t!$ and the displayed powers have their genuine values. Soundness will not assume these identities prematurely.

The pinned CRT construction chooses the representatives at most N^2 and then replaces only a by $a + N^2$. Thus every representative is at most $2N^2$, and their sum is at most $10N^2$. Since $B \geq 2$,

$$N \geq 4^R, \qquad \frac{q}{N^2} = N^{2R-2}, \qquad N^2 \geq 6.$$

Each lower index $k \in \{t, t+1, y+1, x+1\}$ is at most $R-1$. For an ordinary representative and $0 \leq j < k$,

$$\binom{n}{j} \leq \max(1, n^j) \leq (10N^2)^{R-2} < N^{2R-2} = q/N^2,$$

because $10^{R-2} < N^2$, already implied by $N^2 \geq 16^R$.

For the factorial upper index $n_0 = F^2 + t < R^2 + R$, use $R^2 + R \leq 2^{2R}$, $t \leq R$, and $R \geq 2$:

$$\binom{n_0}{j} \leq n_0^t \leq 2^{2R^2} \leq 2^{4R^2-4R} \leq N^{2R-2} = q/N^2 \qquad (0 \leq j < t).$$

For the derived label, the retained literal constructor bound gives

$$J \leq N^{3\ell}, \quad \ell = 2^{22}, \qquad 3\ell(t+1) \leq 2t! = 2F.$$

The latter inequality holds at $t \geq 16$ in the inherited padded construction. Consequently, for $0 \leq j < t+1$,

$$\binom{J}{j} \leq \max(1, J^j) \leq N^{3\ell t} \leq N^{2F} \leq N^{2R-2} = q/N^2.$$

No group has more than five members, and q/N^2 is an integer. Therefore $5(q/N^2) \leq q - 1$. Lemma 4.1 constructs all four new slacks without enlarging the radix or changing the CRT modulus.

5 One natural row authenticates the Pell indices

For $A \geq 2$, define the Pell sequences by

$$X_0 = 1,\ Y_0 = 0, \qquad X_{j+1} = AX_j + (A^2-1)Y_j, \quad Y_{j+1} = X_j + AY_j.$$

Natural norm-one solutions at discriminant A^2-1 have ordinates $Y_j(A)$. The elementary recurrence and multiplication law give

$$Y_j \equiv j \pmod{A-1}, \quad Y_j \geq j, \quad X_j > (A-1)Y_j\ (j \geq 1), \quad Y_{j+1} > (2A-1)Y_j,$$

and $Y_{2j} = 2X_jY_j$. Classification can be proved by descent using the inverse of the smallest positive unit $A + \sqrt{A^2-1}$.

The inherited frame gives, before interpreting any binomial,

$$W \geq 3, \quad A - 1 > W^3, \quad 2 \leq H < W, \quad 1 \leq k_i < W, \quad r = 14 \leq W - 2.$$

A complete retained indexed certificate fixes $V = Y_H(A)$, where the semantic anchor is $H = J$. The other pairs are represented by consecutive ordinates satisfying $Y_i^2 + Z_i^2 - 2AY_iZ_i = 1$; retained multiplication gates keep the literal residuals quadratic. Their output conditions choose the required orientation, as proved in the inherited source.

Theorem 5.1 (Fused natural Pell cap)**.** *Under the frame assumptions above, suppose every Y_i is a natural Pell ordinate and $Q \in \mathbb{N}$ satisfies*

$$\boxed{V = H + \sum_{i=1}^{r} (Y_i - k_i)^2 + (A-1)Q.} \tag{3}$$

Then $Y_i = Y_{k_i}(A)$ for all i. Conversely, canonical pairs extend to a natural Q whenever $H \geq 2\max_i k_i + 1$.

Proof. Write $Y_i = Y_{j_i}(A)$. Equation (3) gives $|Y_i - k_i| \leq \sqrt{V}$. Since $H \geq 2$, $V \geq 2A > 2W$ and $V > 4$. Hence

$$Y_i \leq k_i + \sqrt{V} < W + \sqrt{V} < V.$$

Strict monotonicity implies $j_i < H < W$. Reduction modulo $A-1$ now shows that $A-1$ divides $E_0 = \sum_i (j_i - k_i)^2$. But

$$0 \leq E_0 \leq r(W-1)^2 < W^3 < A - 1.$$

Thus $E_0 = 0$, proving every requested index.

For the converse put $m = \max_i k_i$, and take canonical pairs. Then $E_1 = \sum_i (Y_{k_i} - k_i)^2 \leq rY_m^2$. The doubled-gap assumption and Pell growth give

$$\begin{aligned} V &\geq Y_{2m+1} > (2A-1)Y_{2m} \\ &= 2(2A-1)X_mY_m > 2(2A-1)(A-1)Y_m^2. \end{aligned}$$

Since $A > W$ and $r, H < W$, this is greater than $rY_m^2 + H$. Also $V - H$ and every $Y_{k_i} - k_i$ are divisible by $A-1$. Therefore $Q = (V - H - E_1)/(A-1)$ is a nonnegative integer and satisfies (3). □

5.1 The semantic anchor already has the required gap

The inherited constructive choice gives $N^2 < a \le 2N^2$. Hence

$$H = J \ge 2a^2 > 2N^4.$$

Every other CRT seed is at most N^2; the five non-CRT indices $t, K, y+1, x+1, F^2+t$ are at most N. Thus $m \le 2N^2$. For $N \ge 2$, $2N^4 \ge 4N^2 + 1$, so the required $H \ge 2m+1$ follows with no new coordinate or inequality.

This gap is used only in constructive completeness. Arbitrary zeros need not satisfy the specific CRT representative choice $a > N^2$; the soundness half of Theorem 5.1 does not assume it.

5.2 The two witness directions in their proper order

From a new natural zero, first recover the raw frame bounds. The fused row supplies $H \le V$, so the retained full anchor criterion applies. Theorem 5.1 then identifies all fourteen seed ordinates. Restore the old congruence quotient by

$$Q_{\mathrm{old}} = \frac{\sum_i (Y_i - k_i)^2}{A - 1}.$$

It is integral and nonnegative. Restore the old aggregate slack by $s = Y_H - H - 1 - \sum_i Y_i$. Since $k_i < H$ and $A > W$, growth gives $Y_H - rY_{H-1} > H$, so this slack is natural as well. Finally restore every individual digit slack by Lemma 4.1. One has now obtained a zero of the entire inherited graph; its existing soundness theorem produces the finite refutation.

For completeness, start from the derivation, choose the inherited padded CRT witnesses and the same lift $a \mapsto a + N^2$, and compute the true arithmetic quantities. Section 4 supplies the four digit slacks. Choose the independently bounded Pell parameter in the retained soluble common-root congruence class. Canonical seed pairs and the anchor at J, together with Theorem 5.1, supply Q. All remaining witnesses are unchanged. This is an equivalence of *existence*; it is not a claimed bijection on arbitrary old natural witnesses.

5.3 Literal contraction to 226 variables

The actual source transformation starts with 344 natural coordinates and 293 quadratic rows. Removing obsolete cap sum gates, fusing the cap, and replacing eleven digit bounds by four leaves 332 coordinates and 281 rows. A checked monic projection removes 106 further coordinates and rows, giving 226 coordinates and 175 quadratic residuals. The sparse projection removes 105 and gives 227 coordinates and 176 rows.

Every deleted coordinate has an explicit polynomial restoration certified nonnegative on the natural domain by nonnegative monomials or rational sum-of-squares certificates. The final equations are exactly

$$P_{226}(\mathbf{z}) = \sum_{j=0}^{174} Q_j(\mathbf{z})^2 = 0, \qquad P_{227}(\mathbf{z}) = \sum_{j=0}^{175} \widetilde{Q}_j(\mathbf{z})^2 = 0.$$

All residuals and every collected coefficient are supplied. The principal residual list is printed in Appendix A. There are no variable exponents, factorial symbols, bounded universal quantifiers, or free coefficient parameters in these literal polynomials.

An optional integer-domain companion substitutes

$$z_i = a_i^2 + b_i^2 + c_i(c_i + 1), \qquad 0 \le i < 226.$$

This expression is nonnegative at every integer triple and represents every natural number: apply the three-square theorem to $4z_i + 1$, which is never of the excluded form $4^e(8h + 7)$, then place the unique odd square in the third position. The result is a specified degree-eight polynomial in 678 unrestricted integers. It is a domain conversion, not an improvement on the specialized signed quartics.

6 Ten-positive-variable equations of degrees 323 and 318

This smaller contribution retains package 133's complete four-coefficient tuple and native calibration. It is separate from the direct C222 route. The only variables are

$$J, F, \alpha, i, T, w, Y, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The fixed positive integers β, K_0, M_C, M_F are reconstructed by the finite 943-tile recipe retained in the compact component and printed in Appendix B. In particular $\beta \geq 31$ is odd and $0 < M_C < \beta < M_F < 2\beta$.

Define the following polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & h &= q - 1, & C &= q - F - \alpha,\\
X &= wq, & \mu &= q^2 + 1, & k &= \eta + \zeta,\\
c &= kY + \eta, & A &= \mu Y(X + 1) + 1, & \Delta &= A^2 - 1,\\
H &= 4A - 5, & V_0 &= X\mu^2 Y^2, & E &= 1 + V_0(V_0 + 1)k^2,\\
U &= ic^2, & a &= \Delta U^2 E, & F_0 &= 1 + a,\\
G &= EF_0, & F_s &= 1 + 4aF_0, & Q &= \Delta(F_s - 1).
\end{aligned}$$

Also put

$$\begin{aligned}
R &= [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J,\\
e &= k - R - 1, & B &= (K_0 + w)C - F,\\
b &= c(T + \alpha) - R, & V &= F_s b - c,\\
K_+ &= X(3c - 2X) + 2, & P &= q^3 h, \qquad M = P\mu^2 H.
\end{aligned}$$

The new lower center is

$$d = (A - 1)UE,$$

and the carrier is

$$\begin{aligned}
Z = P\mu^2\{H[n + V(d - bB)] + K_+\}\\
+ H\{\mu^2(hY - q^3 B) - Pe\}.
\end{aligned}$$

The single ordinary polynomial equation is

$$\boxed{Q(Z^2 - M^2 G V^2) + (M^2 G - Z^2) = 0.} \tag{4}$$

No symbol in these abbreviations adds a quantified unknown.

6.1 The shear preserves positive solvability

The 85-operation parent uses the carrier center $n - bVB$. Let its polynomial be N_{85}, and let (4) be N_{87}. The literal ring identity is

$$N_{87}(\text{other variables}, n) = N_{85}(\text{other variables}, n + dV).$$

This alone does not prove positive-domain equivalence: V must first be oriented. The parent's pretransport arguments apply to any rational carrier. From a positive zero they recover positive integers g, y with

$$G = g^2, \qquad QV^2 - (Q-1)y^2 = 1,$$

and prove $b > c$, hence $V > 0$, before using the literal center. These unchanged arguments use the identities

$$F_s = (1+2a)^2, \qquad Q = 4\Delta^2 U^2 G,$$

which make G coprime to $(Q-1)(QV^2-1)$. Valuations recover the square factors and the retained adjacent-Pell descent recovers integrality. Their full ordinary proof is retained unchanged. Since $d > 0$, the new zero therefore gives a positive old coordinate $n + dV$, so inherited soundness applies.

For completeness use the canonical old witnesses. In the parent's proof notation let $s = Y/q^3 > 0$, $\kappa = K_+/H > 0$, $\rho = e/\mu^2 \geq 0$, $t_0 = B/(q-1) \geq 0$, and $W_0 = 1 + (q-1)bV$. These are integers on those witnesses, with

$$n_{85} = gy + W_0 t_0 + \rho - \kappa - s.$$

The exact identity

$$G - d^2 = E + 2(A-1)U^2E^2 > 0 \tag{5}$$

gives the integer bound $g - d \geq 1$. The canonical inequalities give $y > V > 2c$ and $\kappa < c$, $s < c$. Thus

$$\begin{aligned} n_{85} - dV &= (g-d)V + g(y-V) + W_0 t_0 + \rho - \kappa - s \\ &> V - \kappa - s > 0. \end{aligned}$$

This constructs the new positive coordinate and proves the converse.

6.2 Exact arithmetic cost and degree

Both $A-1$ and UE are already live in the parent circuit. Replace its three gates for $bV, bVB, n - bVB$ by five gates for $d, bB, d - bB, V(d-bB), n + V(d-bB)$. The exact count is therefore 87: 52 multiplications, 20 additions, and 15 subtractions. Fixed scalar products and squares each count as a multiplication.

Let Z_0 be the old uncentered numerator, so $Z = MdV + Z_0$. Then the output equals

$$\begin{aligned} &- M^2\{Q(G-d^2) + d^2\}V^2 + 2(Q-1)MdVZ_0 \\ &\qquad + (Q-1)Z_0^2 + M^2G. \end{aligned}$$

The prefix degrees are

$$\deg Q = 86, \quad \deg M = 13, \quad \deg G = 56, \quad \deg V = 80, \quad \deg d = 28, \quad \deg Z_0 \leq 99.$$

Equation (5) has exact degree 51. The four displayed terms have degree at most 323, 306, 284, and 82, respectively; the first has a nonzero highest homogeneous part. With $q_0 = \beta J$, it is

$$-2^{11} q_0^{153} w^{33} Y^{77} i^{14} (\eta + \zeta)^{44} (q_0 - F)^2.$$

For example the coefficient of $J^{153} w^{33} Y^{77} i^{14} \eta^{44} F^2$ is $-2^{11}\beta^{153} \neq 0$. The literal polynomial therefore has exact total degree 323, not merely a syntactic upper bound.

This gives a modest new operations/degree tradeoff between the existing 85/328 and 89/288 points. It does not dominate either of them. The 84/436 parent remains the cheaper evaluator among this small set.

6.3 A sharper center gives 88 operations and degree 318

There is a second valid choice, $d_\sharp = AUE$, in exactly the same carrier and output formula. Compute it by adding the already live UE to $(A-1)UE$, so it costs one additional addition. The count is 88 operations: 52 multiplications, 21 additions, and 15 subtractions. The shear and new-to-old positivity argument are unchanged.

The lower-center inverse bound $g > d$ does not apply to this choice. Instead the inherited canonical source witnesses have $X = 2^R$, $R > 3q$, $w = X/q$, $C \geq 1$, and $F < q$. For $q \geq 32$, this gives $w > q > F$, and therefore $B = (K_0 + w)C - F \geq 1$. The retained pretransport bound is $b > 4g$. Also

$$4G - d_\sharp^2 = 4E + (3A^2 - 4)U^2E^2 > 0,$$

so $d_\sharp < 2g$ and $bB - d_\sharp > 0$. Rewriting the same old canonical witness gives

$$n_{85} - d_\sharp V = gy + t_0 + \rho - \kappa - s + V(bB - d_\sharp) > V - \kappa - s > 0.$$

Thus constructive positive completeness also holds for the sharper center. This uses actual calibrated witnesses, not an assumption that the old carrier coordinate can be freely enlarged.

Its exact defect is $G - d_\sharp^2 = E - U^2E^2$, of degree 46. The four degree bounds in the preceding output identity become 318, 306, 284, and 82. The highest homogeneous part is

$$2^{10}(\beta J)^{150}w^{32}Y^{76}i^{14}(\eta + \zeta)^{44}(\beta J - F)^2,$$

which is nonzero. Hence the total degree is exactly 318. The full component proof further establishes that both shears are bijections on the respective positive zero sets, including either radical-sign branch; the existential directions above already suffice for Theorem 1.1.

7 Calibration, proof scope, and reproducibility

7.1 What is proved here and what is imported

There are three distinct proof layers.

1. **New unconditional finite algebra.** The graph theorem proves an explicit integer zero-set bijection for any checked source system. The local equality lemma and the exact graph identities do not depend on a consistency assumption or a proof calculus.
2. **New unbounded arithmetic.** The shared digit-margin proof, the natural fused cap, their witness order, and the compact lower-center argument are ordinary mathematical proofs. Their essential unbounded claims are not inferred from finite tests.
3. **Inherited consistency calibration.** The signed324 source comes from package 131's direct C222 arithmetic theorem; signed442 comes from package 125. The natural route imports package 129's semantic graph and its calibration. The named logical interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The compact branch retains package 133's separate native/K215 calibration.

The direct source represents a finite syntactic derivation in the numerical calculus C222. Its retained soundness and completeness use factorial fixed points, binomial digits, CRT representatives and Horn derivation decoding. The arithmetic system has a zero exactly when the calculus has a derivation; the two named logical interfaces then connect that derivation to a finite ZFC refutation. These are the precise mathematical dependencies of Theorem 1.1.

No new complete Lean/Rocq development or encoded object-language ZFC proof term is supplied. The existing formal dependency closures were not all rebuilt during this work. This is an ordinary proof formalizable in ZFC, with independently checked finite polynomial artifacts. As in standard mathematical reuse, importing a named source theorem is distinct from claiming to have re-proved its entire foundation. The archive retains source identities, relevant proof texts and explicit boundaries.

7.2 What the exact checks establish

The graph verifier is independent of the producer. It checks triangular normalization, exact restoration of every original residual, complete alias and unit partitions, all logical block sums, the internal spanning trees, and the entire collected coefficient list. It also checks the symbolic canonical lift in (2). Deliberate mutations of source rows, incidence signs, alias labels, free flows and the final constant are rejected. Fresh rebuilds reproduce the mathematical JSON artifacts byte for byte.

The direct checker independently reconstructs the changed raw rows, including the four pooled digit bounds and the fused cap. It retains the unchanged factorial output bound, checks the entire affine export, replays every monic projection with its nonnegativity certificate, and compares every expanded quartic coefficient. A separate reviewer checked the unbounded digit estimates, anchor gap and witness order.

The compact checker reconstructs all four fixed integers from the finite printed tile recipe, checks their hashes and bit lengths, verifies the literal circuit modification and cost, and checks the degree certificate. Its finite-field coefficient controls supplement the displayed symbolic upper bound; they are not the proof of positive solvability.

Integrity hashes identify delivered bytes; they are not theorem-prover certificates or claims of an external signature. “Independent review” here means a separately tasked AI-assisted review and separately written checkers, not an external human endorsement.

7.3 How to use the archive

The full deliverable is self-contained for the supported finite reconstruction and coefficient checks. Python 3.12 is the tested runtime; the implementation uses the standard library. From the archive root:

```
python3 -B verify_all.py --rebuild
python3 -B -O verify_all.py
./build_report.sh
```

The PDF rebuild uses a normal pdfLaTeX installation. PDF bytes may differ across TeX installations; byte equality is required for mathematical data, not for cross-version PDF metadata. The top-level manifest records the delivered files. Component READMEs identify smaller commands and exact artifact formats.

Every graph monomial is an explicit product of indexed coordinates with coefficient 1 or -1. Every direct residual is an explicit integer quadratic. Every compact register is a polynomial abbreviation in the ten declared unknowns and four fixed reconstructed integers. No delivered equation asks the reader to select an unspecified polynomial by DPRM.

8 Further research questions

Joint graph layout and alias optimization. The search first selects a graph and then identifies compatible aliases. Can a joint optimization of edge incidence, opposite-occurrence matching and alias classes reduce the 951-variable endpoint substantially? The complete coefficient certificate permits much more freedom than a global Sidon or four-cycle restriction. A useful objective should record both dimension and support, since fewer hubs can increase monomial count.

Equality penalties and the actual energy. The four-term quartic is locally optimal in the narrow class proved here. It need not be optimal as part of a larger nonnegative polynomial. Can cancellations with the graph energy eliminate the remaining dummy coordinates while preserving height one? Any proposal must certify the sum's nonnegativity and its zero set, not just its coefficient list.

Pool more arithmetic inequalities. The direct gain came from proving that several canonical remainders jointly fit below one denominator. Other source bounds may have comparable unused margins. Which can be pooled without a circular dependence on the factorial or binomial interpretation? A sufficient condition must be proved for canonical witnesses, and a literal restoration map must handle every new zero.

Transfer the pooled digits to the signed source. The current 226-variable reduction is over $\mathbb{N}$. Signed systems recover remainder signs from product guards. Sharing their bounds may remove more variables, but requires a fresh sign argument; the natural-domain proof does not automatically transfer. A smaller signed source would also improve the height-one compiler's output.

Improve the compact center. The center $(A-1)UE$ cancels five leading degrees at a cost of two gates. Can another polynomial lower center reduce the degree more while preserving the explicit positive reconstruction? A formal coordinate shear is not enough: positivity of the transformed witness must be proved in both directions and in the correct order.

Close the formal calibration chain. The graph transformation is a small, general target for proof-assistant formalization: finite maps, integer polynomial identities, positive semidefinite energies, and unimodular tree elimination. The direct arithmetic bridge is more substantial. A future formal release should make the exact internal-ZFC endpoint explicit rather than treating finite script checks as a substitute.

Meaningful lower bounds. The only optimality statement here is the local two-variable equality gadget theorem. The global minimum number of variables, degree, support, coefficient height or operations for a Con(ZFC) equation remains unknown. A valuable next step would be a lower bound for a precisely specified compiler class, not an unsupported claim that one displayed equation is the simplest possible.

A Complete principal natural residuals

The following is the complete 175-row quadratic specification of $P_{226} = \sum_j Q_j^2$. The variables are $z_0, \dots, z_{225} \in \mathbb{N}$. Rows are generated directly from the authoritative equation JSON and checked against it. The collected expansion is also included in the archive; the row format is substantially shorter to read.

$$Q_0 = -z_2^2 + z_3$$

$$Q_1 = 1 - 2z_2 - z_2z_4 + z_4 + z_5 - z_6$$

$$Q_2 = 1 + z_4 - z_6 - z_7$$

$$Q_3 = -17 - z_0 - z_0z_2 - z_0z_9 - 17z_2 - 17z_9 + z_{11}$$

$$
\begin{aligned}
Q_{4} = & -1+z_{1}-z_{2}-z_{2}z_{8}-2z_{8}-z_{8}z_{9}\\
Q_{5} = & -1+z_{1}-z_{10}-z_{10}z_{11}\\
Q_{6} = & 2+z_{3}-z_{4}z_{22}+z_{20}-2z_{22}\\
Q_{7} = & -6-5z_{12}-z_{12}^{2}-2z_{12}z_{13}-5z_{13}-z_{13}^{2}\\
& +z_{23}\\
Q_{8} = & -72-68z_{13}-16z_{13}^{2}-16z_{13}z_{23}-34z_{23}-4z_{23}^{2}\\
& +z_{24}\\
Q_{9} = & -30-11z_{12}-z_{12}^{2}-4z_{12}z_{24}-22z_{24}-4z_{24}^{2}\\
& +z_{25}\\
Q_{10} = & -380-312z_{24}-64z_{24}^{2}-32z_{24}z_{25}-78z_{25}-4z_{25}^{2}\\
& +z_{26}\\
Q_{11} = & -6-5z_{13}-z_{13}^{2}-2z_{13}z_{14}-5z_{14}-z_{14}^{2}\\
& +z_{27}\\
Q_{12} = & -6-5z_{12}-z_{12}^{2}-2z_{12}z_{14}-5z_{14}-z_{14}^{2}\\
& +z_{28}\\
Q_{13} = & -414-22z_{12}-2z_{12}^{2}-16z_{12}z_{14}-8z_{12}z_{27}-200z_{13}\\
& -32z_{13}^{2}-64z_{13}z_{14}-32z_{13}z_{23}-32z_{13}z_{28}-320z_{14}-64z_{14}^{2}\\
& -32z_{14}z_{23}-32z_{14}z_{27}-32z_{14}z_{28}-100z_{23}-8z_{23}^{2}-16z_{23}z_{28}\\
& -52z_{27}-8z_{27}^{2}-108z_{28}-8z_{28}^{2}+z_{29}\\
Q_{14} = & -1344-800z_{13}-128z_{13}^{2}-256z_{13}z_{14}-128z_{13}z_{23}-128z_{13}z_{28}\\
& -864z_{14}-128z_{14}^{2}-128z_{14}z_{23}-128z_{14}z_{28}-400z_{23}-32z_{23}^{2}\\
& -64z_{23}z_{28}-432z_{28}-32z_{28}^{2}-2z_{29}-2z_{29}^{2}+z_{30}\\
Q_{15} = & -247-168z_{12}-32z_{12}^{2}-64z_{12}z_{13}-32z_{12}z_{27}-32z_{12}z_{28}\\
& -188z_{13}-32z_{13}^{2}-32z_{13}z_{27}-32z_{13}z_{28}-2z_{23}-92z_{27}\\
& -8z_{27}^{2}-16z_{27}z_{28}-84z_{28}-8z_{28}^{2}+z_{31}\\
Q_{16} = & -960-672z_{12}-128z_{12}^{2}-256z_{12}z_{13}-128z_{12}z_{27}-128z_{12}z_{28}\\
& -736z_{13}-128z_{13}^{2}-128z_{13}z_{27}-128z_{13}z_{28}-368z_{27}-32z_{27}^{2}\\
& -64z_{27}z_{28}-336z_{28}-32z_{28}^{2}-2z_{31}-2z_{31}^{2}+z_{32}\\
Q_{17} = & -276-30z_{12}-2z_{12}^{2}-16z_{12}z_{14}-8z_{12}z_{27}-30z_{13}\\
& -2z_{13}^{2}-16z_{13}z_{14}-8z_{13}z_{28}-272z_{14}-64z_{14}^{2}-32z_{14}z_{27}\\
& -32z_{14}z_{28}-68z_{27}-8z_{27}^{2}-68z_{28}-8z_{28}^{2}+z_{33}\\
Q_{18} = & -552-120z_{13}-8z_{13}^{2}-64z_{13}z_{14}-32z_{13}z_{28}-544z_{14}\\
& -128z_{14}^{2}-128z_{14}z_{28}-272z_{28}-32z_{28}^{2}-2z_{33}-2z_{33}^{2}\\
& +z_{34}\\
Q_{19} = & -247-4z_{12}-352z_{13}-128z_{13}^{2}-64z_{13}z_{23}-64z_{13}z_{27}\\
& -92z_{23}-8z_{23}^{2}-16z_{23}z_{27}-84z_{27}-8z_{27}^{2}-2z_{28}\\
& +z_{35}\\
Q_{20} = & -960-1408z_{13}-512z_{13}^{2}-256z_{13}z_{23}-256z_{13}z_{27}-368z_{23}\\
& -32z_{23}^{2}-64z_{23}z_{27}-336z_{27}-32z_{27}^{2}-2z_{35}-2z_{35}^{2}\\
& +z_{36}
\end{aligned}
$$

$$
\begin{aligned}
Q_{21} = &-56-60z_{13}-16z_{13}^2-16z_{13}z_{23}-30z_{23}-4z_{23}^2\\
&+z_{37}\\
Q_{22} = &-30-11z_{12}-z_{12}^2-4z_{12}z_{37}-22z_{37}-4z_{37}^2\\
&+z_{38}\\
Q_{23} = &-380-312z_{37}-64z_{37}^2-32z_{37}z_{38}-78z_{38}-4z_{38}^2\\
&+z_{39}\\
Q_{24} = &-30-11z_{13}-z_{13}^2-4z_{13}z_{39}-22z_{39}-4z_{39}^2\\
&+z_{40}\\
Q_{25} = &-349-420z_{13}-128z_{13}^2-128z_{13}z_{23}-208z_{23}-32z_{23}^2\\
&-2z_{27}+z_{41}\\
Q_{26} = &-1368-1664z_{13}-512z_{13}^2-512z_{13}z_{23}-832z_{23}-128z_{23}^2\\
&-2z_{41}-2z_{41}^2+z_{42}\\
Q_{27} = &-56-60z_{12}-16z_{12}^2-16z_{12}z_{28}-8z_{12}z_{42}-30z_{28}\\
&-4z_{28}^2-4z_{28}z_{42}-15z_{42}-z_{42}^2+z_{43}\\
Q_{28} = &-6-5z_{14}-z_{14}^2-2z_{14}z_{15}-5z_{15}-z_{15}^2\\
&+z_{44}\\
Q_{29} = &-224-26z_{12}-2z_{12}^2-16z_{12}z_{15}-8z_{12}z_{44}-288z_{14}\\
&-128z_{14}^2-64z_{14}z_{27}-64z_{14}z_{28}-88z_{15}-32z_{15}^2-32z_{15}z_{44}\\
&-76z_{27}-8z_{27}^2-16z_{27}z_{28}-68z_{28}-8z_{28}^2-44z_{44}\\
&-8z_{44}^2+z_{45}\\
Q_{30} = &-640-1152z_{14}-512z_{14}^2-256z_{14}z_{27}-256z_{14}z_{28}-304z_{27}\\
&-32z_{27}^2-64z_{27}z_{28}-272z_{28}-32z_{28}^2-2z_{45}-2z_{45}^2\\
&+z_{46}\\
Q_{31} = &-241-168z_{12}-32z_{12}^2-64z_{12}z_{13}-32z_{12}z_{27}-32z_{12}z_{28}\\
&-184z_{13}-32z_{13}^2-32z_{13}z_{27}-32z_{13}z_{28}-z_{14}-92z_{27}\\
&-8z_{27}^2-16z_{27}z_{28}-84z_{28}-8z_{28}^2+z_{47}\\
Q_{32} = &-42-52z_{14}-16z_{14}^2-16z_{14}z_{27}-26z_{27}-4z_{27}^2\\
&+z_{48}\\
Q_{33} = &-30-11z_{14}-z_{14}^2-4z_{14}z_{48}-22z_{48}-4z_{48}^2\\
&+z_{49}\\
Q_{34} = &-306-35z_{13}-z_{13}^2-16z_{13}z_{48}-4z_{13}z_{49}-280z_{48}\\
&-64z_{48}^2-32z_{48}z_{49}-70z_{49}-4z_{49}^2+z_{50}\\
Q_{35} = &-4556-4320z_{48}-1024z_{48}^2-512z_{48}z_{49}-128z_{48}z_{50}-1080z_{49}\\
&-64z_{49}^2-32z_{49}z_{50}-270z_{50}-4z_{50}^2+z_{51}\\
Q_{36} = &-30-11z_{14}-z_{14}^2-4z_{14}z_{51}-22z_{51}-4z_{51}^2\\
&+z_{52}\\
Q_{37} = &-306-35z_{13}-z_{13}^2-16z_{13}z_{51}-4z_{13}z_{52}-280z_{51}\\
&-64z_{51}^2-32z_{51}z_{52}-70z_{52}-4z_{52}^2+z_{53}
\end{aligned}
$$

$$
\begin{aligned}
Q_{38} = &-1028-672z_{12}-128z_{12}^2-256z_{12}z_{13}-128z_{12}z_{27}-128z_{12}z_{28}\\
&-736z_{13}-128z_{13}^2-128z_{13}z_{27}-128z_{13}z_{28}-368z_{27}-32z_{27}^2\\
&-64z_{27}z_{28}-336z_{28}-32z_{28}^2-2z_{47}-2z_{47}^2-32z_{51}\\
&-8z_{52}-2z_{53}+z_{54}
\end{aligned}
$$

$$
\begin{aligned}
Q_{39} = &-264-128z_{51}-32z_{52}-8z_{53}-2z_{54}-2z_{54}^2\\
&+z_{55}
\end{aligned}
$$

$$
\begin{aligned}
Q_{40} = &-2-3z_{12}-z_{12}^2-2z_{12}z_{55}-3z_{55}-z_{55}^2\\
&+z_{56}
\end{aligned}
$$

$$
\begin{aligned}
Q_{41} = &-12-28z_{55}-16z_{55}^2-16z_{55}z_{56}-14z_{56}-4z_{56}^2\\
&+z_{57}
\end{aligned}
$$

$$
\begin{aligned}
Q_{42} = &-30-11z_{14}-z_{14}^2-4z_{14}z_{57}-22z_{57}-4z_{57}^2\\
&+z_{58}
\end{aligned}
$$

$$
\begin{aligned}
Q_{43} = &-306-35z_{12}-z_{12}^2-16z_{12}z_{57}-4z_{12}z_{58}-280z_{57}\\
&-64z_{57}^2-32z_{57}z_{58}-70z_{58}-4z_{58}^2+z_{59}
\end{aligned}
$$

$$
\begin{aligned}
Q_{44} = &-4556-4320z_{57}-1024z_{57}^2-512z_{57}z_{58}-128z_{57}z_{59}-1080z_{58}\\
&-64z_{58}^2-32z_{58}z_{59}-270z_{59}-4z_{59}^2+z_{60}
\end{aligned}
$$

$$
\begin{aligned}
Q_{45} = &-90-76z_{14}-16z_{14}^2-16z_{14}z_{27}-16z_{14}z_{60}-38z_{27}\\
&-4z_{27}^2-8z_{27}z_{60}-38z_{60}-4z_{60}^2+z_{61}
\end{aligned}
$$

$$
\begin{aligned}
Q_{46} = &-4556-4320z_{51}-1024z_{51}^2-512z_{51}z_{52}-128z_{51}z_{53}-1080z_{52}\\
&-64z_{52}^2-32z_{52}z_{53}-270z_{53}-4z_{53}^2+z_{62}
\end{aligned}
$$

$$
\begin{aligned}
Q_{47} = &-682-70z_{13}-2z_{13}^2-32z_{13}z_{60}-8z_{13}z_{61}-592z_{60}\\
&-128z_{60}^2-64z_{60}z_{61}-148z_{61}-8z_{61}^2-2z_{62}+z_{63}
\end{aligned}
$$

$$
Q_{48} = -z_{63}-z_{63}^2+z_{64}
$$

$$
\begin{aligned}
Q_{49} = &-42-52z_{13}-16z_{13}^2-8z_{13}z_{14}-16z_{13}z_{23}-13z_{14}\\
&-z_{14}^2-4z_{14}z_{23}-26z_{23}-4z_{23}^2+z_{65}
\end{aligned}
$$

$$
\begin{aligned}
Q_{50} = &-2278-2264z_{13}-800z_{13}^2-800z_{13}z_{23}-160z_{13}z_{65}-70z_{14}\\
&-2z_{14}^2-32z_{14}z_{62}-8z_{14}z_{64}-1132z_{23}-200z_{23}^2-80z_{23}z_{65}\\
&-592z_{62}-128z_{62}^2-64z_{62}z_{64}-148z_{64}-8z_{64}^2-228z_{65}\\
&-8z_{65}^2+z_{66}
\end{aligned}
$$

$$
\begin{aligned}
Q_{51} = &-2712-280z_{14}-8z_{14}^2-128z_{14}z_{62}-32z_{14}z_{64}-2368z_{62}\\
&-512z_{62}^2-256z_{62}z_{64}-592z_{64}-32z_{64}^2-2z_{66}-2z_{66}^2\\
&+z_{67}
\end{aligned}
$$

$$
Q_{52} = -2-3z_{67}-z_{67}^2+z_{68}
$$

$$
Q_{53} = -z_2-z_2z_{17}+z_{69}
$$

$$
Q_{54} = -z_{16}^2+2z_{16}z_{17}-z_{17}^2+z_{70}
$$

$$
Q_{55} = -2z_{26}z_{69}-4z_{69}+z_{71}
$$

$$
\begin{aligned}
Q_{56} = &-z_{16}^2+2z_{16}z_{17}+2z_{16}z_{71}-z_{17}^2-2z_{17}z_{71}-z_{71}^2\\
&+z_{72}
\end{aligned}
$$

$$
Q_{57} = -z_{30}z_{69}+z_{73}
$$

$$Q_{58} = -z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{73} - z_{17}^2 - 2z_{17}z_{73} - z_{73}^2 + z_{74}$$

$$Q_{59} = -z_{32}z_{69} + z_{75}$$

$$Q_{60} = -z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{75} - z_{17}^2 - 2z_{17}z_{75} - z_{75}^2 + z_{76}$$

$$Q_{61} = -z_{34}z_{69} + z_{77}$$

$$Q_{62} = -z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{77} - z_{17}^2 - 2z_{17}z_{77} - z_{77}^2 + z_{78}$$

$$Q_{63} = -z_{36}z_{69} + z_{79}$$

$$Q_{64} = -z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{79} - z_{17}^2 - 2z_{17}z_{79} - z_{79}^2 + z_{80}$$

$$Q_{65} = -4z_{13}z_{69} - 2z_{40}z_{69} - 4z_{69} + z_{81}$$

$$Q_{66} = -z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{81} - z_{17}^2 - 2z_{17}z_{81} - z_{81}^2 + z_{82}$$

$$Q_{67} = -4z_{42}z_{69} - 2z_{43}z_{69} + z_{83}$$

$$Q_{68} = -z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{83} - z_{17}^2 - 2z_{17}z_{83} - z_{83}^2 + z_{84}$$

$$Q_{69} = -z_{46}z_{69} + z_{85}$$

$$Q_{70} = -z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{85} - z_{17}^2 - 2z_{17}z_{85} - z_{85}^2 + z_{86}$$

$$Q_{71} = -z_{12}^2 + 2z_{12}z_{14} - z_{14}^2 + z_{87}$$

$$Q_{72} = -z_{13}^2 + 2z_{13}z_{14} - z_{14}^2 + z_{88}$$

$$Q_{73} = -z_{87}z_{88} + z_{89}$$

$$Q_{74} = -4 - 4z_{15} - z_{15}^2 + z_{90}$$

$$Q_{75} = -z_{89}^2 + 2z_{89}z_{90} - z_{90}^2 + z_{91}$$

$$Q_{76} = -2z_{68}z_{69} - 4z_{69} + z_{92}$$

$$Q_{77} = -z_{16}^2 + 2z_{16}z_{17} + 2z_{16}z_{92} - z_{17}^2 - 2z_{17}z_{92} - z_{92}^2 + z_{93}$$

$$Q_{78} = -z_2 - z_2z_{13} + z_{94}$$

$$Q_{79} = -1 - z_{17} - z_{17}z_{94} - z_{94} + z_{95}$$

$$Q_{80} = -z_2 - z_2z_{12} + z_{96}$$

$$Q_{81} = -4z_2 - 4z_2z_{13} - 2z_2z_{23} + z_{97}$$

$$Q_{82} = -1 - z_{16} - z_{16}z_{96} - z_{96} + z_{98}$$

$$Q_{83} = -z_{97}z_{98} - z_{98} + z_{99}$$

$$Q_{84} = -z_{95}^2 + 2z_{95}z_{99} - z_{99}^2 + z_{100}$$

$$Q_{85} = -6z_2 - 4z_2z_{13} - 2z_2z_{23} + z_{101}$$

$$Q_{86} = -1 - z_{17} - z_{17}z_{101} - z_{101} + z_{102}$$

$$Q_{87} = -1 - z_{16} - z_{16}z_{94} - z_{94} + z_{103}$$

$$
\begin{aligned}
Q_{88} &= -z_{102}^2 + 2z_{102}z_{103} - z_{103}^2 + z_{104} \\
Q_{89} &= -z_{70}z_{72} + z_{105} \\
Q_{90} &= -z_{74}z_{105} + z_{106} \\
Q_{91} &= -z_{76}z_{106} + z_{107} \\
Q_{92} &= -z_{78}z_{107} + z_{108} \\
Q_{93} &= -z_{80}z_{108} + z_{109} \\
Q_{94} &= -z_{82}z_{109} + z_{110} \\
Q_{95} &= -z_{84}z_{110} + z_{111} \\
Q_{96} &= -z_{86}z_{111} + z_{112} \\
Q_{97} &= -z_{91}z_{112} - z_{93}z_{112} + z_{113} \\
Q_{98} &= -z_{100}z_{113} + z_{114} \\
Q_{99} &= -2 - 2z_2 - z_2z_{20} - 2z_9 - z_9z_{20} - z_{20} \\
&\quad + z_{115} \\
Q_{100} &= -1 - z_{18} - z_{18}z_{115} - z_{115} + z_{116} \\
Q_{101} &= -2 - z_2 - z_2z_{19} - z_9 - z_9z_{19} - 2z_{19} \\
&\quad - z_{19}z_{115} - z_{115} + z_{117} \\
Q_{102} &= 2 - 4z_1 + 2z_1^2 - 2z_1z_{16} - 2z_1z_{17} - 2z_1z_{116} \\
&\quad - 2z_1z_{117} + 2z_{16} + z_{16}^2 + 2z_{16}z_{116} + 2z_{17} + z_{17}^2 \\
&\quad + 2z_{17}z_{117} - z_{21}z_{118} + z_{104}z_{114} + 2z_{116} + z_{116}^2 + 2z_{117} \\
&\quad + z_{117}^2 \\
Q_{103} &= -9 - z_{12} - z_{13} - z_{14} - z_{15} - z_{16} \\
&\quad - z_{17} - z_{18} - z_{19} - z_{20} + 10z_{21}^2 - z_{119} \\
Q_{104} &= -2 - z_4 - z_4z_{169} - z_4z_{170} - z_4z_{221} - 3z_{169} \\
&\quad - z_{169}z_{171} - 3z_{170} - z_{170}z_{171} - z_{171} - z_{171}z_{221} + z_{176} \\
&\quad + z_{177} + z_{183} + z_{188} + z_{193} - 3z_{221} + z_{224} \\
Q_{105} &= -132 - 4z_0 - 4z_0z_{122} - 3z_1 - 3z_1z_{122} - 5z_2 \\
&\quad - 5z_2z_{122} - 2z_3 - 2z_3z_{122} - 2z_4 - 2z_4z_{122} - z_5 \\
&\quad - z_5z_{122} - 3z_9 - 3z_9z_{122} - 2z_{12} - 2z_{12}z_{122} - 14z_{13} \\
&\quad - 14z_{13}z_{122} - z_{14} - z_{14}z_{122} - z_{15} - z_{15}z_{122} - z_{16} \\
&\quad - z_{16}z_{122} - z_{17} - z_{17}z_{122} - z_{18} - z_{18}z_{122} - z_{19} \\
&\quad - z_{19}z_{122} - z_{20} - z_{20}z_{122} - z_{21} - z_{21}z_{122} - 4z_{23} \\
&\quad - 4z_{23}z_{122} - 2z_{26} - 2z_{26}z_{122} - z_{30} - z_{30}z_{122} - z_{32} \\
&\quad - z_{32}z_{122} - z_{34} - z_{34}z_{122} - z_{36} - z_{36}z_{122} - 2z_{40} \\
&\quad - 2z_{40}z_{122} - 4z_{42} - 4z_{42}z_{122} - 2z_{43} - 2z_{43}z_{122} - z_{46} \\
&\quad - z_{46}z_{122} - 2z_{68} - 2z_{68}z_{122} - 132z_{122} - z_{122}z_{169} - z_{122}z_{170} \\
&\quad - z_{122}z_{171} - z_{122}z_{198} - z_{122}z_{199} - z_{122}z_{205} - z_{122}z_{210} - z_{122}z_{211} \\
&\quad - z_{122}z_{216} - z_{122}z_{221} + z_{123} - z_{169} - z_{170} - z_{171} \\
&\quad - z_{198} - z_{199} - z_{205} - z_{210} - z_{211} - z_{216} \\
&\quad - z_{221}
\end{aligned}
$$

$$
\begin{aligned}
Q_{106} = & -4z_0z_{123} - 3z_1z_{123} - 5z_2z_{123} - 2z_3z_{123} - 2z_4z_{123} - z_5z_{123} \\
& - 3z_9z_{123} - 2z_{12}z_{123} - 14z_{13}z_{123} - z_{14}z_{123} - z_{15}z_{123} - z_{16}z_{123} \\
& - z_{17}z_{123} - z_{18}z_{123} - z_{19}z_{123} - z_{20}z_{123} - z_{21}z_{123} - 4z_{23}z_{123} \\
& - 2z_{26}z_{123} - z_{30}z_{123} - z_{32}z_{123} - z_{34}z_{123} - z_{36}z_{123} - 2z_{40}z_{123} \\
& - 4z_{42}z_{123} - 2z_{43}z_{123} - z_{46}z_{123} - 2z_{68}z_{123} - 133z_{123} - z_{123}z_{169} \\
& - z_{123}z_{170} - z_{123}z_{171} - z_{123}z_{198} - z_{123}z_{199} - z_{123}z_{205} - z_{123}z_{210} \\
& - z_{123}z_{211} - z_{123}z_{216} - z_{123}z_{221} + z_{124}
\end{aligned}
$$

$$
Q_{107} = -1 + z_{120}^2 + z_{123}^2 - z_{124}^2
$$

$$
\begin{aligned}
Q_{108} = & -17 - 8z_4 - z_4^2 + 2z_4z_{120} + 2z_4z_{121} - 2z_4z_{171} \\
& + 8z_{120} + 2z_{120}z_{171} + 8z_{121} + 2z_{121}z_{171} - z_{125}z_{126} - 8z_{171} \\
& - z_{171}^2
\end{aligned}
$$

$$
Q_{109} = -z_{21} - z_{21}z_{210} - z_{21}z_{211} - z_{21}z_{216} + z_{126}
$$

$$
Q_{110} = -4z_{128} - 4z_{128}z_{133} + z_{134}
$$

$$
Q_{111} = -z_{120}z_{128} - z_{121}z_{128} + z_{135}
$$

$$
Q_{112} = -1 + z_{127}^2 + z_{128}^2 - z_{135}^2
$$

$$
Q_{113} = -z_{120}z_{130} - z_{121}z_{130} + z_{136}
$$

$$
Q_{114} = -1 + z_{129}^2 + z_{130}^2 - z_{136}^2
$$

$$
Q_{115} = -z_{132} - z_{132}z_{134} + z_{137}
$$

$$
Q_{116} = -1 + z_{131}^2 + z_{132}^2 - z_{137}^2
$$

$$
Q_{117} = -z_{128}^2 + z_{139}
$$

$$
Q_{118} = z_{130} - z_{138}z_{139} - z_{139}
$$

$$
Q_{119} = 1 - z_{120} - z_{121} - z_{129}z_{140} + z_{134}
$$

$$
Q_{120} = -z_{127} - z_{129}z_{141} + z_{131}
$$

$$
\begin{aligned}
Q_{121} = & -25 - z_{12} - 13z_{13} - 4z_{23} - 2z_{26} - z_{30} \\
& - z_{32} - z_{34} - z_{36} - 2z_{40} - 4z_{42} - 2z_{43} \\
& - z_{46} - 2z_{68} - 4z_{128}z_{142} + z_{132}
\end{aligned}
$$

$$
Q_{122} = -z_{143}z_{144} + z_{145}
$$

$$
Q_{123} = -1 - 2z_{120}z_{145} - 2z_{121}z_{145} + z_{143}^2 + z_{144}^2
$$

$$
\begin{aligned}
Q_{124} = & -2 - 2z_3 - z_3^2 + 2z_3z_{120} + 2z_3z_{121} + 2z_{120} \\
& + 2z_{121} - z_{146}
\end{aligned}
$$

$$
Q_{125} = z_3z_{143} - z_5 + z_{143} - z_{144} - z_{146}z_{147}
$$

$$
Q_{126} = -z_{148}z_{149} + z_{150}
$$

$$
Q_{127} = -1 - 2z_{120}z_{150} - 2z_{121}z_{150} + z_{148}^2 + z_{149}^2
$$

$$
\begin{aligned}
Q_{128} = & -5 - 4z_4 - z_4^2 + 2z_4z_{120} + 2z_4z_{121} + 4z_{120} \\
& + 4z_{121} - z_{151}
\end{aligned}
$$

$$
Q_{129} = z_4z_{148} - z_{21} + 2z_{148} - z_{149} - z_{151}z_{152}
$$

$$
Q_{130} = -1 - z_{21}^2 + 2z_{21}z_{120} + 2z_{21}z_{121} - z_{153}
$$

$$
Q_{131} = -3 - z_4 + z_{21}z_{148} - z_{149} - z_{153}z_{154} - z_{171}
$$

$$
\begin{aligned}
Q_{132} = &-10 - 6z_4 - z_4^2 + 2z_4z_{120} + 2z_4z_{121} - 2z_4z_{171} \\
&+ 6z_{120} + 2z_{120}z_{171} + 6z_{121} + 2z_{121}z_{171} - z_{155} - 6z_{171} \\
&- z_{171}^2 \\
Q_{133} = &-1 + z_4z_{143} + 3z_{143} + z_{143}z_{171} - z_{144} - z_{155}z_{156} \\
&- z_{169} - z_{170} - z_{221} \\
Q_{134} = &-z_{157}z_{158} + z_{159} \\
Q_{135} = &-1 - 2z_{120}z_{159} - 2z_{121}z_{159} + z_{157}^2 + z_{158}^2 \\
Q_{136} = &-1 + z_4z_{157} - z_{155}z_{160} + 3z_{157} + z_{157}z_{171} - z_{158} \\
&- z_{198} - z_{199} - z_{205} \\
Q_{137} = &-z_{161}z_{162} + z_{163} \\
Q_{138} = &-1 - 2z_{120}z_{163} - 2z_{121}z_{163} + z_{161}^2 + z_{162}^2 \\
Q_{139} = &-1 + z_4z_{161} - z_{155}z_{164} + 3z_{161} + z_{161}z_{171} - z_{162} \\
&- z_{210} - z_{211} - z_{216} \\
Q_{140} = &-z_{165}z_{166} + z_{167} \\
Q_{141} = &-1 - 2z_{120}z_{167} - 2z_{121}z_{167} + z_{165}^2 + z_{166}^2 \\
Q_{142} = &-2 - z_4 + z_4z_{165} - z_4z_{169} - z_4z_{170} - z_4z_{221} \\
&+ 4z_{165} + z_{165}z_{171} - z_{166} - z_{168} - z_{168}z_{176} - z_{168}z_{177} \\
&- z_{168}z_{183} - z_{168}z_{188} - z_{168}z_{193} - z_{168}z_{224} - 3z_{169} - 2z_{170} \\
&- 2z_{221} \\
Q_{143} = &-z_{172}z_{173} + z_{174} \\
Q_{144} = &-1 - 2z_{120}z_{174} - 2z_{121}z_{174} + z_{172}^2 + z_{173}^2 \\
Q_{145} = &-z_{21} - z_{21}z_{176} - z_{21}z_{177} - z_{21}z_{183} - z_{21}z_{188} - z_{21}z_{193} \\
&- z_{21}z_{224} + z_{178} \\
Q_{146} = &\, z_4z_{172} + z_{171}z_{172} + 4z_{172} - z_{173} - z_{175}z_{178} - z_{176} \\
Q_{147} = &-z_{179}z_{180} + z_{181} \\
Q_{148} = &-1 - 2z_{120}z_{181} - 2z_{121}z_{181} + z_{179}^2 + z_{180}^2 \\
Q_{149} = &\, z_4z_{179} + z_{171}z_{179} - z_{178}z_{182} + 4z_{179} - z_{180} - z_{183} \\
Q_{150} = &-z_{184}z_{185} + z_{186} \\
Q_{151} = &-1 - 2z_{120}z_{186} - 2z_{121}z_{186} + z_{184}^2 + z_{185}^2 \\
Q_{152} = &\, z_4z_{184} + z_{171}z_{184} - z_{178}z_{187} + 4z_{184} - z_{185} - z_{188} \\
Q_{153} = &-z_{189}z_{190} + z_{191} \\
Q_{154} = &-1 - 2z_{120}z_{191} - 2z_{121}z_{191} + z_{189}^2 + z_{190}^2 \\
Q_{155} = &\, z_4z_{189} + z_{171}z_{189} - z_{178}z_{192} + 4z_{189} - z_{190} - z_{193} \\
Q_{156} = &-z_{194}z_{195} + z_{196} \\
Q_{157} = &-1 - 2z_{120}z_{196} - 2z_{121}z_{196} + z_{194}^2 + z_{195}^2 \\
Q_{158} = &-z_{21} - z_{21}z_{198} - z_{21}z_{199} - z_{21}z_{205} + z_{200} \\
Q_{159} = &\, z_4z_{194} + z_{171}z_{194} + 4z_{194} - z_{195} - z_{197}z_{200} - z_{198} \\
Q_{160} = &-z_{201}z_{202} + z_{203}
\end{aligned}
$$

$$
\begin{aligned}
Q_{161} &= -1 - 2z_{120}z_{203} - 2z_{121}z_{203} + z_{201}^2 + z_{202}^2 \\
Q_{162} &= z_4 z_{201} + z_{171}z_{201} - z_{200}z_{204} + 4z_{201} - z_{202} - z_{205} \\
Q_{163} &= -z_{206}z_{207} + z_{208} \\
Q_{164} &= -1 - 2z_{120}z_{208} - 2z_{121}z_{208} + z_{206}^2 + z_{207}^2 \\
Q_{165} &= z_4 z_{206} - z_{126}z_{209} + z_{171}z_{206} + 4z_{206} - z_{207} - z_{210} \\
Q_{166} &= -z_{212}z_{213} + z_{214} \\
Q_{167} &= -1 - 2z_{120}z_{214} - 2z_{121}z_{214} + z_{212}^2 + z_{213}^2 \\
Q_{168} &= z_4 z_{212} - z_{126}z_{215} + z_{171}z_{212} + 4z_{212} - z_{213} - z_{216} \\
Q_{169} &= -z_{217}z_{218} + z_{219} \\
Q_{170} &= -1 - 2z_{120}z_{219} - 2z_{121}z_{219} + z_{217}^2 + z_{218}^2 \\
Q_{171} &= -z_{21} - z_{21}z_{169} - z_{21}z_{170} - z_{21}z_{221} + z_{222} \\
Q_{172} &= z_4 z_{217} + z_{171}z_{217} + 4z_{217} - z_{218} - z_{220}z_{222} - z_{221} \\
Q_{173} &= z_4 z_{128} + z_{127} + 4z_{128} + z_{128}z_{171} - z_{135} - z_{178}z_{223} \\
&\quad - z_{224} \\
Q_{174} &= 1995 + 216z_0 + 6z_0^2 + 8z_0z_1 + 16z_0z_2 + 2z_0z_3 \\
&\quad + 8z_0z_9 - 2z_0z_{143} - 4z_0z_{148} - 2z_0z_{165} + 154z_1 + 5z_1^2 \\
&\quad + 16z_1z_2 + 8z_1z_9 - 4z_1z_{148} - 2z_1z_{161} + 308z_2 + 17z_2^2 \\
&\quad + 18z_2z_9 - 8z_2z_{148} - 2z_2z_{157} + 32z_3 + z_3^2 - 2z_3z_{165} \\
&\quad + 156z_9 + 5z_9^2 - 4z_9z_{148} - 2z_9z_{157} + 3z_{12} + z_{12}^2 \\
&\quad - 2z_{12}z_{172} + 15z_{13} + z_{13}^2 - 2z_{13}z_{179} + 2z_{14} + z_{14}^2 \\
&\quad - 2z_{14}z_{184} + 2z_{15} + z_{15}^2 - 2z_{15}z_{189} + 2z_{16} + z_{16}^2 \\
&\quad - 2z_{16}z_{194} + 2z_{17} + z_{17}^2 - 2z_{17}z_{201} + 2z_{18} + z_{18}^2 \\
&\quad - 2z_{18}z_{206} + 2z_{19} + z_{19}^2 - 2z_{19}z_{212} + 2z_{20} + z_{20}^2 \\
&\quad - 2z_{20}z_{217} + 4z_{23} + 2z_{26} + z_{30} + z_{32} + z_{34} \\
&\quad + z_{36} + 2z_{40} + 4z_{42} + 2z_{43} + z_{46} + 2z_{68} \\
&\quad + z_{120}z_{225} + z_{121}z_{225} - z_{128} - 32z_{143} + z_{143}^2 - 76z_{148} \\
&\quad + z_{148}^2 - 4z_{157} + z_{157}^2 - 2z_{161} + z_{161}^2 - 32z_{165} \\
&\quad + z_{165}^2 - 2z_{172} + z_{172}^2 - 2z_{179} + z_{179}^2 - 2z_{184} \\
&\quad + z_{184}^2 - 2z_{189} + z_{189}^2 - 2z_{194} + z_{194}^2 - 2z_{201} \\
&\quad + z_{201}^2 - 2z_{206} + z_{206}^2 - 2z_{212} + z_{212}^2 - 2z_{217} \\
&\quad + z_{217}^2 - z_{225}
\end{aligned}
$$

B Complete compact circuit and fixed coefficients

The register table below gives all 87 paid binary operations of the lower-center equation. For the sharp-center equation replace the single row `d=Am1*UE` by `d0=Am1*UE` followed by `d=d0+UE`; every other row is unchanged. Its output is the polynomial (4); the register names introduce no new unknowns. The four ports are fixed by the finite recipe following the table. The recipe is transcribed from the pinned complete parent display, and its full four-integer output is independently bound by hashes and bit lengths in the circuit artifact.

Step	Register	Binary definition
1	bJ	beta * J
2	q	bJ + 1
3	qmF	q - F
4	C	qmF - alpha
5	X	w * q
6	k	eta + zeta
7	kY	k * Y
8	c	kY + eta
9	q2	q * q
10	mu	q2 + 1
11	Xp1	X + 1
12	muY	mu * Y
13	Am1	muY * Xp1
14	4Am1	4 * Am1
15	Hp	4Am1 - 1
16	Ap1	Am1 + 2
17	muY2	muY * muY
18	V0	X * muY2
19	V0k	V0 * k
20	V0kk	V0k + k
21	V0k2	V0k * V0kk
22	Dtau	V0k2 + 1
23	G1	bJ * qmF
24	G	G1 + alpha
25	q2m1	q2 - 1
26	Gq	G * q2m1
27	qMF	q * MF
28	MCqMF	MC + qMF
29	RJ	MCqMF * J
30	R	Gq + RJ
31	q3	q2 * q
32	kR	k - R
33	e	kR - 1
34	K0w	K0 + w
35	K0wC	K0w * C
36	Bt	K0wC - F
37	c2	c * c
38	U	i * c2
39	UE	U * Dtau
40	Delta	Am1 * Ap1
41	SE	Delta * UE
42	DpI2	SE * U
43	Df	DpI2 + 1
44	aF0	DpI2 * Df
45	fouraF0	4 * aF0
46	Fsqr	fouraF0 + 1
47	Q	Delta * fouraF0
48	Tp	T + alpha
49	cT	c * Tp
50	cTR	cT - R
51	DfcTR	Fsqr * cTR
52	V	DfcTR - c
53	V2	V * V
54	P1	q3 * bJ

Step	Register	Binary definition
55	mu2	mu * mu
56	P1mu2	P1 * mu2
57	d	Am1 * UE
58	bB	cTR * Bt
59	dmbB	d - bB
60	Vcenter	V * dmbB
61	center	n + Vcenter
62	Hn	Hp * center
63	threec	3 * c
64	twoX	2 * X
65	gap	threec - twoX
66	Kpos0	X * gap
67	Kpos	Kpos0 + 2
68	lin	Hn + Kpos
69	Zpart1	P1mu2 * lin
70	qm1Y	bJ * Y
71	q3Bt	q3 * Bt
72	inner	qm1Y - q3Bt
73	mu2inner	mu2 * inner
74	P1e	P1 * e
75	brk	mu2inner - P1e
76	Zpart2	Hp * brk
77	Znum	Zpart1 + Zpart2
78	Mtot	P1mu2 * Hp
79	M2	Mtot * Mtot
80	Dg	Dtau * Df
81	Z2	Znum * Znum
82	RG	M2 * Dg
83	RGV2	RG * V2
84	inside	Z2 - RGV2
85	Qinside	Q * inside
86	correction	RG - Z2
87	N	Qinside + correction

The following finite Python recipe defines the four integers without a repository checkout or a network request. Every range is fixed; no input variable of the Diophantine equation occurs in it. The surrounding source file also checks the resulting byte hashes against the pinned parent.

```
"""Reconstruct the complete indexed BFS943 tuple printed by package 133.

No parameters remain after this finite integer computation.  The formulas
are transcribed from the pinned parent_printed_equation.tex, not inferred
from the numerical hashes.
"""
from pathlib import Path
import hashlib
import json

ALPHABET = "0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz"
ENCODED = (
    "BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y"
    "iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U"
    "OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq"
    "4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ"
    "FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F"
    "We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl"
    "0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ"
    "8P"
```

```
)

def construct():
    if len(ENCODED) != 436:
        raise ValueError("incorrect action numeral length")
    number = 0
    for c in ENCODED:
        number = 62 * number + ALPHABET.index(c)
    N = number % (1 << 636)
    T0 = (number >> 636) % (1 << 1675)
    E0 = number >> 2311
    nu = [(N >> j) & 1 for j in range(430)]
    m = [(N >> (430+q)) & 1 for q in range(215)]
    actions = []
    for q in range(215):
        fq = 14 * (1-m[q]) + m[q] * ((E0 >> (4 * sum(m[:q]))) & 15)
        for b in range(2):
            j = 2*q+b
            sigma = sum(nu[:j])
            u = nu[j] * (1+sigma) + (1-nu[j]) * ((T0 // (216 ** (j-sigma))) % 216)
            v = (fq >> (2*b)) & 1
            d = (fq >> (2*b+1)) & 1
            actions.append((u,v,d))
    right = {u for u,v,d in actions if d == 1}
    left = {u for u,v,d in actions if d == 0}
    rp = {q:6+sum(x<q for x in right) for q in right}
    lp = {q:205+sum(x<q for x in left) for q in left}
    tiles = {}
    def put(j,t):
        if j in tiles:
            raise ValueError("duplicate tile index")
        tiles[j]=t
    h=lambda q,b:2*(q+1)+b
    put(0,(2,435,3,1)); put(1,(5,0,0,0))
    for j,(u,v,d) in enumerate(actions):
        put(j+2,((1-d)*lp.get(u,0),h(j//2,j%2),d*rp.get(u,0),v))
    for b in range(2):
        put(432+b,(0,h(215,b),0,h(215,b)))
        put(434+b,(0,b,0,b))
    for q in right:
        for b in range(2): put(424+2*rp[q]+b,(rp[q],b,0,h(q,b)))
    for q in left:
        for b in range(2): put(424+2*lp[q]+b,(0,b,lp[q],h(q,b)))
    for j in range(4): put(924+j,(0,j+430*(j//2),0,438))
    put(928,(2,434,2,0)); put(929,(3,436,4,4)); put(930,(4,436,4,0))
    for t in (434,435,436): put(497+t,(1,438,1,t))
    for a in (0,1,4):
        for b in range(3): put(934+3*a+b-6*(a//4),(a,437,b,437))
    if set(tiles) != set(range(943)):
        raise ValueError("incomplete indexed tiles")
    rho=lambda e:1 << (67*e)
    B=rho(8807)
    positions=(set(range(943))|set(range(983,993))|set(range(1003,1012))|
               set(range(1013,1032))|{1063,1064,1095,2195,3296})
    MC=B-1-sum(rho(e) for e in positions-{1})-2*rho(1063)
    MF=(B-1+4+((1<<51)-2)*(rho(4393)+rho(5495))+
        sum(rho(1013+j) for j in range(10))+sum(rho(1065+j) for j in range(9))+
        rho(2195)+rho(3296)+((1<<65)-1)*sum(rho(8792+j) for j in range(13))+
        ((1<<19)-1)*rho(8805))
    K=sum(rho(e) for e in (30,62,1100,2201,8849))
    for base in (4393,5495):
        for j in range(943):
            w,n,e,s=tiles[j]
            value=w+(n<<8)+(e<<17)+(s<<25)+(127<<34)
            K+=((1+(1<<51))-(1<<10)*value)*rho(base-j)
        for b,e,m_ in ((10,1023,8),(18,1013,9),(27,1003,8),(35,983,9),
                       (44,1031,1),(45,1022,1),(46,1011,1),(47,992,1),
                       (48,1095,1),(49,2195,1),(50,3296,1)):
            K+=sum((1<<(b+j))*rho(base-e-j) for j in range(m_))
    return dict(beta=B-1,K0=K,MC=MC,MF=MF)
```

C Artifact inventory and source references

Path	Purpose
`graph/equations/`	Four literal height-one polynomials, readable expansions, graph certificates and metrics.
`graph/independent_verify.py`	Independent finite algebra, graph and complete coefficient checker.
`graph/source324/`, `graph/source442/`	Exact pinned source equations; signed324 proof and calibration scope.
`direct/equations/`	Natural226 and natural227 residuals, complete expansions, projection certificates and integer678 recipe.
`direct/PROOF.md`	Detailed pooled-digit and fused-cap proof, including witness order.
`compact/`	Literal 87- and 88-operation polynomials, fixed coefficient reconstruction, parent comparison and exact checks.
`verification/`	Final integrated verification receipts and document review.
`PROVENANCE.json`	Inspected repository commit and component source boundaries.
`MANIFEST.sha256`	Integrity inventory of the delivered files.

References

[1] ProveIt, package 127, *Shared arithmetic certificates and graph energies for Con(ZFC)*. Pinned repository package. The inherited graph energy, shared hub construction and sufficient four-cycle condition are predecessors of the present compiler.

[2] ProveIt, package 125, *Round 15 unified binomial and even Pell index equations*. Pinned repository package. Source of the signed442 comparison equation.

[3] ProveIt, package 128, *One-row Pell certificates*. Pinned repository package. Predecessor for combining a cap and an index congruence.

[4] ProveIt, package 129, *A semantic Pell anchor*. Pinned repository package. The source of the semantic label anchor, CRT lift, consecutive ordinates and the natural233/natural234 comparison equations.

[5] ProveIt, package 131, *Shared Pell caps, round 16*. Pinned repository package. Source of the signed324 equation and its ordinary arithmetic calibration.

[6] ProveIt, package 133, *Staged certificates*. Pinned repository package. The 84/436 and 85/328 parents, product-square proof and complete indexed coefficient tuple of the compact component.

[7] J. Bayer, M. David, B. Stock, A. Pal, Y. Matiyasevich and D. Schleicher, *Diophantine Equations and the DPRM Theorem*, Archive of Formal Proofs. `https://isa-afp.org/entries/DPRM_Theorem.html`. General background only; no unspecified DPRM polynomial is used as a delivered equation in this report.